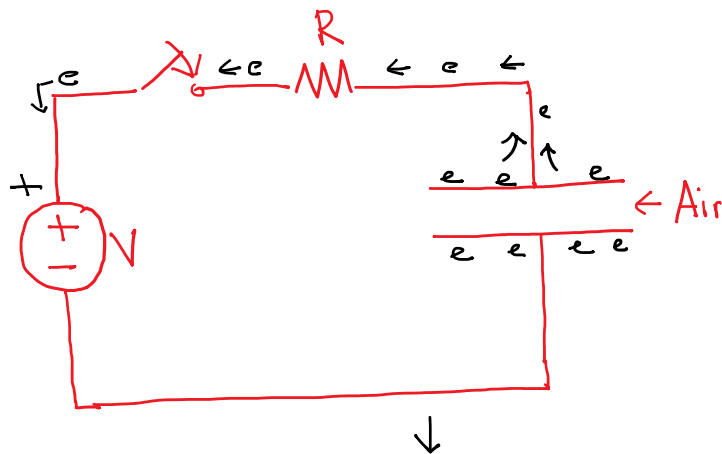


□ Capacitors

□ Inductors

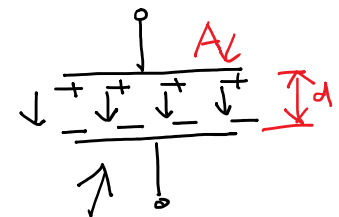
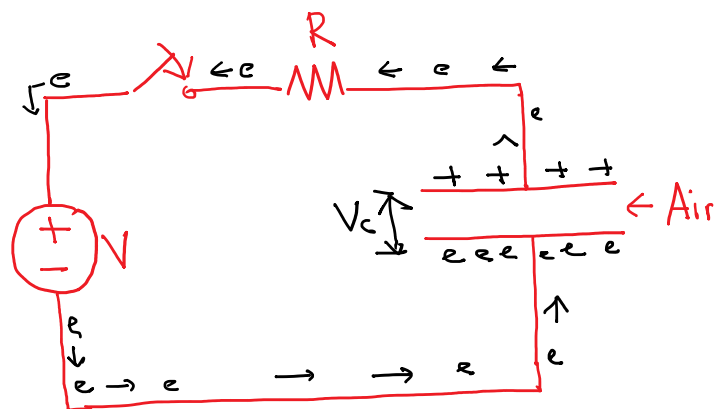


e e e e

e e e e

Metals.

Good Conductor

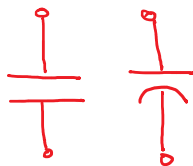
↓
Plenty of free electrons

Store charge

Something that hold charge

$$Q \propto V_c \Rightarrow Q = CV$$

C: Proportionality Constant;
Capacitance: Farade: $\left(\frac{\text{Coulomb}}{\text{Volt}} \right)$



1 Farade, if the capacitor store 1Q charge at 1V

$$C = \frac{\epsilon_r A}{d}$$

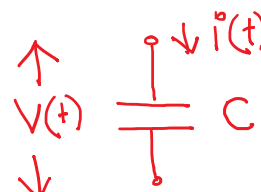
ϵ_r : Relative permittivity
dielectric constant.

$$Q = CV \Rightarrow \frac{dQ}{dt} = C \frac{dV}{dt} \Rightarrow i(t) = C \frac{dV}{dt}$$

$$\frac{dV(t)}{dt} = \frac{1}{C} i(t)$$

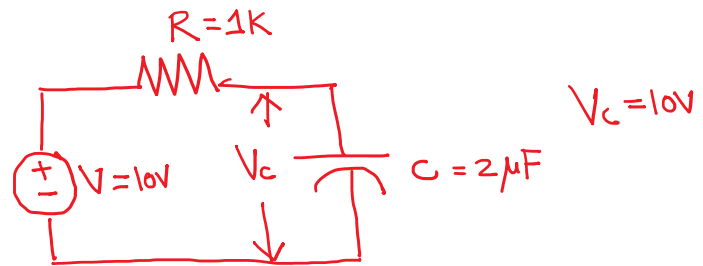
$$\Rightarrow dV(t) = \frac{1}{C} i(t) dt$$

$$V(t) = \frac{1}{C} \int_{t_0}^t i(t) dt + V(t_0)$$

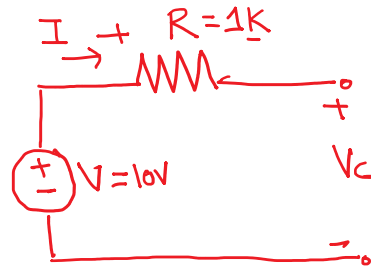


$$i(t) = C \frac{dV}{dt} = C \times 0 = 0$$

$V = DC / \text{Const.}$
 to DC Signal (Voltage/current) Capacitor is always treated as open ckt.

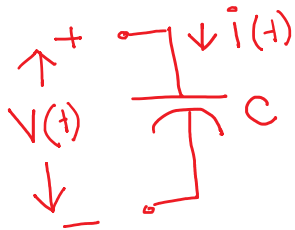


$t < 0: V = 0$

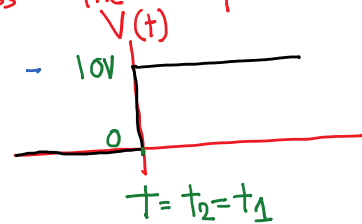


$$\begin{aligned} +V - V_R - V_C &= 0 \\ \Rightarrow 10 - \cancel{I_R} - V_C &= 0 \\ \Rightarrow V_C &= 10V \end{aligned}$$

Voltage across a capacitor doesn't change abruptly/Instantaneously



lets consider the voltage profile across the capacitor is as followed:



at $t = t_2 = t_1$, Voltage changes abruptly from $0V \rightarrow 10V$

at $t = t_1$, $V(t_1) = 0V$

at $t = t_2$, $V(t_2) = 10V$

Q: what is the current $i(t)$??

$$i(t) = C \frac{dV(t)}{dt} = C \frac{V(t_2) - V(t_1)}{t_2 - t_1} = \frac{10 - 0}{0} = \infty \quad (\text{undefined})$$

Summary: $i(t) = \infty$, which is not possible,

Therefore: Voltage across capacitor cannot change abruptly or instantaneously